Neeloy Chakraborty

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Interests: Embodied AI, Generative AI, Robot Teleoperation, Anomaly and Hallucination Detection, Reinforcement Learning

Skills: Python, PyTorch, PyTorch3D, Ollama, OpenCV, Blender, ROS, ZED Camera, Linux, Git, CARLA, Docker, C/C++

Education:

- **University of Illinois Urbana-Champaign**
  PhD in Electrical and Computer Engineering (Exp. May 2026) GPA 3.90/4.00
  Master of Science in Electrical and Computer Engineering (2021 – 2023) [Thesis] GPA 3.83/4.00
  Bachelor of Science in Computer Engineering (2017 – 2021) [Thesis] GPA 3.75/4.00

Select Experience:

- **Research Assistant in the Human-Centered Autonomy Lab** Fall 2019 – present
  Designing intuitive instruction-following embodied AI agents [Website]
  Developing real-time image generation methods combining depth foundation models and rendering techniques
  Detecting failures and hallucinations in LLM generations within multi-agent safety-critical environments

- **PhD Research Intern in Sight Lab at Dolby Laboratories** Summer 2025
  Researching greenfield project within the space of creative 3D asset generation [Website]

- **Research and Advanced Engineering Intern in Core AI/ML at Ford Motor Company** Summer 2022
  Designing sample efficient model-free + model-based RL methods [Website]
  Outperforming classical PID controllers by 41% in complex autonomous vehicle use case

- **Perception Engineering Intern in Autonomy Team at Brunswick i-Jet Lab** Summer 2021
  Localizing swimmers around boats using filtering, tracking, and computer vision techniques [Website]
  Analyzing functional safety standards practiced at company and presenting findings to global management team

- **Engineering Intern in Global CAD Team at Qualcomm** Summer 2020
  Leading design process of generalized data gathering solutions to support Design for Test pipeline [Website]
  Collaborating and adapting with international teams to consider multiple perspectives

Select Publications, Patents, and Pre-prints:

- **N. Chakraborty**, J. Pohovey, M. Ornik, and K. Driggs-Campbell. "Adaptive Stress Testing Black-Box LLM Planners," Under review 2025 [Paper]

- **N. Chakraborty***, Y. Fang*, A. Schreiber, T. Ji, Z. Huang, A. Mihigo, C. Wall, A. Almana, and K. Driggs-Campbell. "Towards Real-Time Generation of Delay-Compensated Video Feeds for Outdoor Mobile Robot Teleoperation," ICRA 2025 [Paper] [Website] [Code] [Dataset] **(Leading cross-departmental team of researchers with diverse skillsets)**

- K. Balakrishnan, **N. Chakraborty**, and D. Upadhyay. "Model-based Reinforcement Learning," US20240320505A1 [Patent Pending]

- **N. Chakraborty***, R. Sidhu*, B. Abdullai*, H. Chen*, N. Ravi*, A. Ankur, D. Prasad, and J. Hockenmaier. "BEAST: Building an Embodied Action-prediction System with Trajectory data," Alexa Prize SimBot Challenge Proceedings 2023 [Paper] [Website] **(University team leader and semi-finalist in inaugural Amazon SimBot Challenge)**

- **N. Chakraborty**, A. Hasan*, S. Liu*, T. Ji*, W. Liang, D. L. McPherson, and K. Driggs-Campbell. "Structural Attention-based Recurrent Variational Autoencoder for Highway Vehicle Anomaly Detection," AAMAS 2023 [Paper] [Website] [Code] **(Accepted as full paper with 23.3% acceptance rate and received student scholarship)**